In [3]:
```
import pandas as  pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
```

In [4]:
```
df = pd.read_csv('house.csv')
df.head()
```

Out[4]:

	House Size	House Price
0	1500	350000
1	1800	400000
2	2000	450000
3	2200	475000
4	2500	525000

In [5]:
```
X = df[['House Size']]
Y = df['House Price']
```

In [6]:
```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2,
```

In [7]:
```
model = LinearRegression()
model.fit(X_train, Y_train)
```

Out[7]:

▾ LinearRegression

Parameters

	fit_intercept	True
	copy_X	True
	tol	1e-06
	n_jobs	None
	positive	False

In [8]:
```
Y_pred = model.predict(X_test)

Y_pred
```

Out[8]:
```
array([ 957571.17449272, 3092324.31941217, 2219016.2146724 ,
        3674529.72257202, 1151639.64221267, 3965632.42415194,
        1830879.2792325 , 1733845.04537252, 2413084.68239235,
        1248673.87607265])
```

In [9]:

```
r2_score = r2_score(Y_test, Y_pred)
r2_score

#Print evaluation metrics
print(f"R-squared Score: ",r2_score)
```

```
R-squared Score:  0.9987913548048982
```

In [10…

```
user_House_Size = int(input("Enter House Size to predict salary: "))

user_df = pd.DataFrame([[user_House_Size]], columns=[ 'House Size'])

predicted_salary = model.predict(user_df)
print (f"The predicted House Price For House Size {user_House_Size} is:",
```

```
The predicted House Price For House Size 1800 is: [336552.07778889]
```

In [12…

```
# Plot
plt.scatter(X, Y, color='blue', label='Data points')  # ALL data points
plt.plot(X, model.predict(X), color='red', label='Regression Line')  # Re
plt.xlabel('House Size (sqft)')
plt.ylabel('House Price ($)')
plt.title('House Price vs. Size')
plt.legend()
plt.show()
```

In [ ]:

```

```